

Department of Mechanical Engineering

Course Outcomes of all courses of B Tech 7th semester MECHANICAL

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C401 Design of Transmission System D037711(037)	C 401.1	Model, analyze and design spur gears. (Level 4)
	C 401.2	Model, analyze and design helical and bevel gears. (Level 4)
	C 401.3	Model, analyze and design springs and brakes. (Level 4)
	C 401.4	Model, analyze and design bearings. (Level 4)
	C 401.5	Model, analyze and design chain and belt drives. (Level 4)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C402- Refrigeration & Air- Conditioning D037712(037)	C402.1	Analyze vapour compression refrigeration system. (Level 4)
	C402.2	Analyze gas and air cycle refrigeration system. (Level 4)
	C402.3	Analyze vapour absorption system, describe refrigerant and refrigeration equipment. (Level 4)
	C402.4	Explain terminologies of psychrometry and human comfort and apply to analyze related problems. (Level 3,4)
	C402.5	Carry out cooling load calculations and describe air-conditioning systems. (Level 2)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C403- Automation in Manufacturing D037713(037)	C403.1	Illustrate the basic concepts of automation in machine. (Level 3)
	C403.2	Explain the fundamentals of CAD. . (Level 2)
	C403.3	Explain the basics of computer aided manufacturing. . (Level 2)
	C403.4	Discuss the low cost automation systems. (Level 4)
	C403.5	Explain the basic concepts of modeling and simulation. (Level 2)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C404-Machine Tool Technology D037731(037)	C404.1	Demonstrate an understanding of cutting tool materials and tool geometries and apply mechanics of metal cutting for analysis of related problems. (Level 3)
	C404.2	Demonstrate an understanding of concepts of machinability, mechanism of tool failure, thermal aspects in machining and cutting fluid. (Level 3)
	C404.3	Describe the construction features of machine tool elements and analyze the forces and torque acting on it. (Level 2)
	C404.4	Design speed gear box. (Level 6)
	C 404.5	Design feed gear box and describe acceptance tests of machine tools. (Level 6)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C405- Non Conventional Energy Sources D000724(025)	C405.1	Demonstrate the generation of electricity from various Non-Conventional sources of energy, have a working knowledge on types of fuel cells. (Level 3)
	C405.2	Estimate the solar energy, Utilization of it, Principles involved in solar energy collection and conversion of it to electricity generation. (Level 3,2,4)
	C405.3	Explore the concepts involved in wind energy conversion system by studying its components, types and performance. (Level 3)
	C405.4	Illustrate ocean energy and explain the operational methods of their utilization. (Level 3)
	C405.5	Acquire the knowledge on geothermal energy. (Level 1)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C406- Refrigeration & Air- Conditioning Lab D037721(037)	C406.1	Describe the construction and working of different refrigeration and air conditioning equipments. (Level 2)
	C406.2	Analyze performance parameters of refrigeration system. (Level 4)
	C406.3	Analyze performance parameters of mechanical heat pump. (Level 4)
	C406.4	Analyze performance parameters of air conditioning system. (Level 4)
	C406.5	Simulate and analyze various air conditioning processes. (Level 3,6,4)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C407- CIM & Automation Lab D037722(037)	C407.1	Demonstrate an understanding of concepts discussed in Computer Integrated Manufacturing course and its implementation in manufacturing (Level 3)
	C407.2	Write CNC part programs using CADEM simulation package for simulation of machining operations such as Turning, Drilling & Milling. (Level 6)
	C407.3	Write programs for Flexible Manufacturing Systems. (Level 6)
	C407.4	Write programs for Robotics. (Level 6)
	C407.5	Demonstrate an understanding of the operating principles of hydraulics, pneumatics and electro–pneumatic systems. (Level 3,2)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C408- Project (Phase I) D037723(037)	C408.1	Technical Skills Development Students will apply programming languages (e.g., Java, Kotlin) and development frameworks (e.g., Android Studio) to build a functional mobile application. Level: 2
	C408.2	Problem-Solving Abilities Students will analyze user requirements and identify challenges in task management. They will devise solutions to address usability issues and enhance user experience through features such as task categorization and prioritization. Level: 3
	C408.3	Communication Skills Enhancement Students will evaluate user feedback obtained through surveys and usability testing. They will communicate their findings effectively through project reports and presentations, articulating improvements made based on user input. Level: 4
	C408.4	Professional Development Students will create a professional portfolio showcasing their mobile application development skills. They will document the project's design process, coding practices, and testing methodologies to demonstrate their proficiency to potential employers. Level: 5
	C408.5	Collaboration and Teamwork Students will apply effective collaboration and teamwork skills throughout the project lifecycle. They will assign roles and responsibilities, communicate project milestones, and resolve conflicts to ensure the timely delivery of the mobile application. Level: 2

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C409- Industrial Training D037724(037)	C409.1	Technical Skill Development Participants will apply programming languages (e.g., Java, Python) and development frameworks (e.g., Spring Boot, Django) to create functional software applications. Level: 3
	C409.2	Problem-Solving Abilities Participants will analyze complex software problems and debug code effectively using problem-solving techniques such as root cause analysis and systematic debugging strategies. Level: 4
	C409.3	Communication and Documentation Participants will evaluate their ability to communicate technical concepts clearly through written documentation, such as technical specifications, user manuals, and project reports, ensuring effective knowledge transfer and collaboration. Level: 5
	C409.4	Project Management Skills Participants will create project plans, including task scheduling, resource allocation, and risk management strategies, demonstrating their ability to manage software development projects effectively. Level: 6
	C409.5	Collaboration and Teamwork Participants will apply effective teamwork and collaboration skills while working on group projects or collaborating with colleagues, fostering a positive team environment and achieving project goals collectively. Level: 2